

Subhadeep Sarkar

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SUMMARY

Computer Science and Engineering undergraduate specializing in AI and Machine Learning. Proficient in Java, Python, and C++ with strong expertise in data structures, algorithms, and software development. Adaptable team player recognized for analytical problem-solving and clear communication of technical concepts.

TECHNICAL SKILLS

Programming: Java, Python, C++

Web Development: HTML, CSS, Basics of JavaScript

Cloud: Amazon Web Services (AWS: S3, ECS, EKS, Fargate, IAM)

Concepts: Artificial Intelligence, Machine Learning, Deep Learning, Data Structures, Algorithms

Other: Problem Solving, Analytical Thinking, Debugging, Collaboration, Communication

Frameworks: Django, Pandas, Pytorch, Matplotlib

DevOps & Tools: Docker, Git, Render, Versal

PROJECTS

Biomedical Knowledge Graph from Literature

2024

Technologies: Python, OpenCV, BERT (Sequence Classification, graph mining)

- Developed a knowledge graph pipeline that extracted and classified over 10,000 unique biomedical relationships from scientific literature, improving discovery of COVID-19-related entity interactions by 30% compared to manual methods. `pyecharts`.
- Automated data preprocessing and relation extraction workflows, reducing manual analysis time by 70% and enabling real-time visualization for faster hypothesis generation among researchers.

Car Price Predictor

2026

Technologies: Python, FastAPI, Scikit-learn, Pandas, NumPy, Flask/Streamlit, Render(Cloud Hosting),Git

- Engineered a real-time forest fire detection system that increased early detection accuracy by 25% using deep learning on drone imagery and IoT sensor data, compared to traditional manual monitoring.
- Reduced alert response time from detection to notification to under 30 seconds (previously 5 minutes), via edge processing and cloud-based dashboards. Enabled scalable remote monitoring over a 10 sq km test site, supporting rapid emergency response and minimizing fire damage.

Program-O Chatbot

2026

Technologies: PHP, MySQL, JavaScript(jQuery), AJAX, HTML5, CSS3

- Designed and built a basic web-based chatbot using PHP and MySQL that instantly answers user messages by looking up pre-defined conversation rules stored in a database.
- Created a clean web page using JavaScript and AJAX so users can type and receive real-time text replies smoothly without the screen reloading.

EDUCATION

Vellore Institute of Technology

B.Tech, Computer Science and Engineering (AI/ML)

– GPA: 8.83/10

July 2023 – May 2027 (expected)

Bhopal, India

CERTIFICATIONS & BADGES

- Applied Machine Learning in Python - University of Michigan (2025)
- AWS Cloud Clubs Machine Learning Camper - AWS Community (2024)
- Prompt Design in Vertex AI Skill Badge - Google Cloud (2025)